Module 5: Self-Organizing Maps

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WATER CONSUMPTION PATTERNS IN CONTINENTAL PORTUGAL (2011-2022)

WHAT IS THE ISSUE?

- **Climate change** projections: rising temperatures, frequent droughts
- National per capita consumption: 163.8L/day (2001) → 171.8L/day (2020)
- Water management requires **regional** understanding

RESEARCH OBJECTIVE

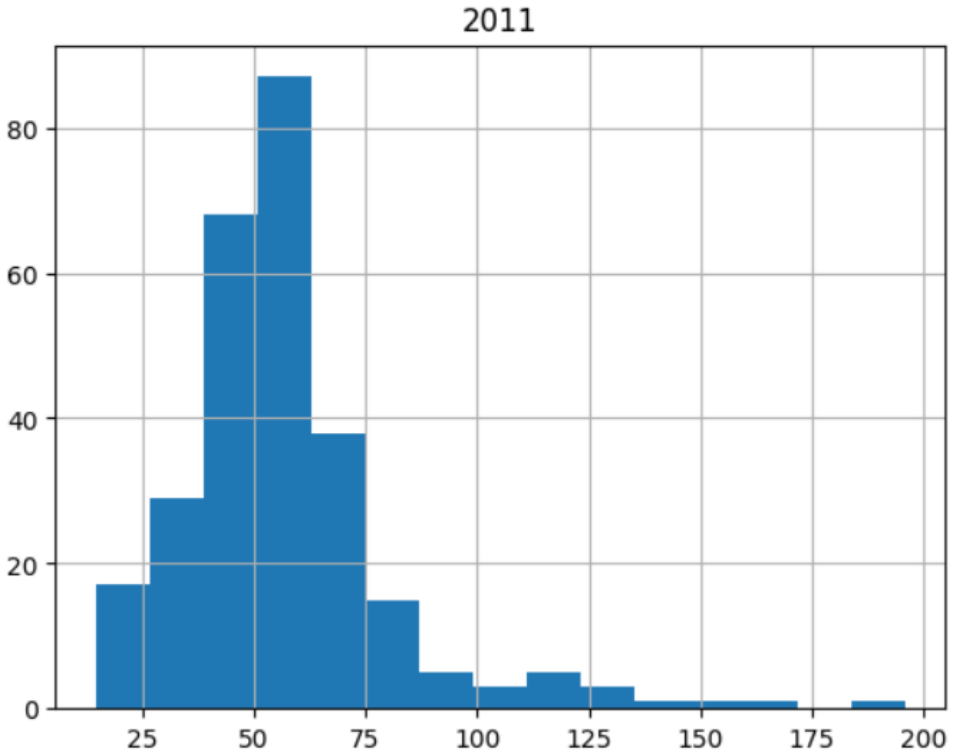
Identify **spatial patterns**: Which municipalities have similar consumption behaviors?

Identify **temporal patterns**: Which years share similar consumption regimes?

DATA AND METHODS

- Data: **274** municipalities and **12** years
  - Raw data: 308 municipalities and 21 years
  - Pre-processing: Data breaks and geographical transformations
  - Source: PORDATA
- Method: Self-Organizing Maps (SOM)
- Metric: Annual water consumed per capita (m^3/hab)
- Missing data: linear interpolation

EXPLORATORY DATA ANALYSIS (EDA)



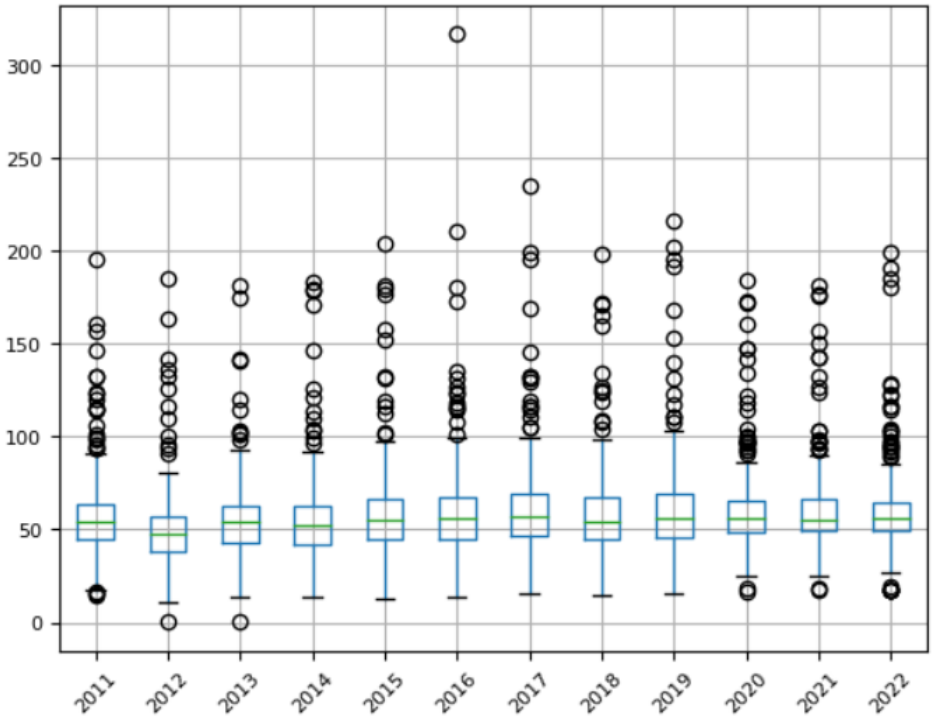
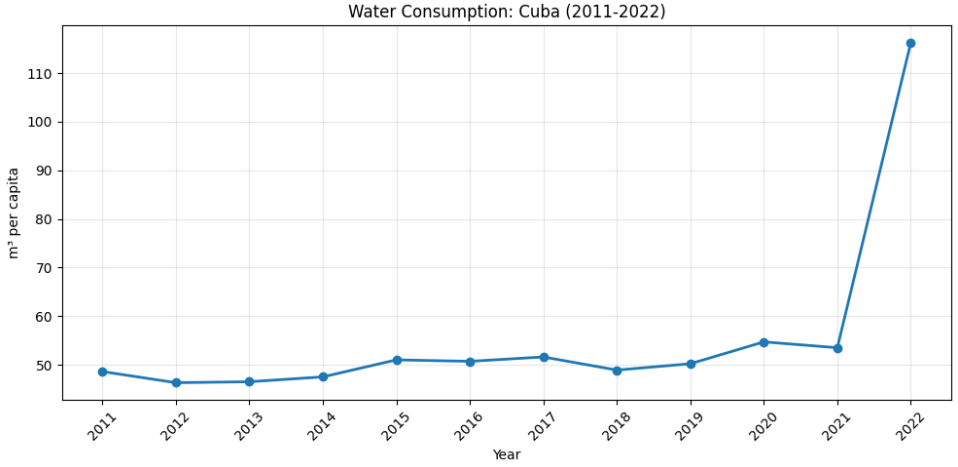


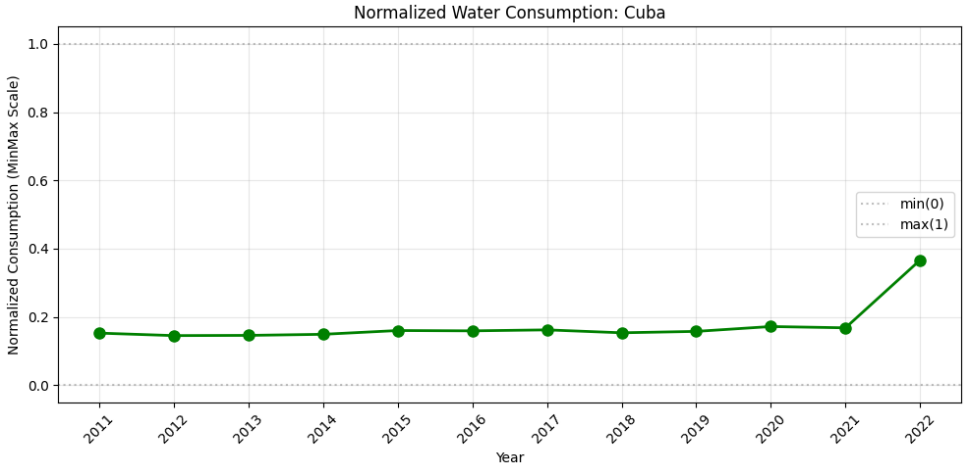
Figure 1 (left): Histogram (2011); Figure 2 (right): Boxplot graph (2011-2022)

EXPLORATORY DATA ANALYSIS (EDA)

Município (Municipality)	Average (2011-2022) m^3/hab
Marco de Canaveses	**15.47**
Cinfães	17.02
Paredes	18.3
Soure	55.3
Cuba	**55.5**
Sobral de Monte Agraço	55.6
Vila do Bispo	170.28
Lagoa	177.76
Albufeira	**196.47**

Table 1 (above): Top 3 highest, middle 3, bottom 3 lowest consumption municipalities





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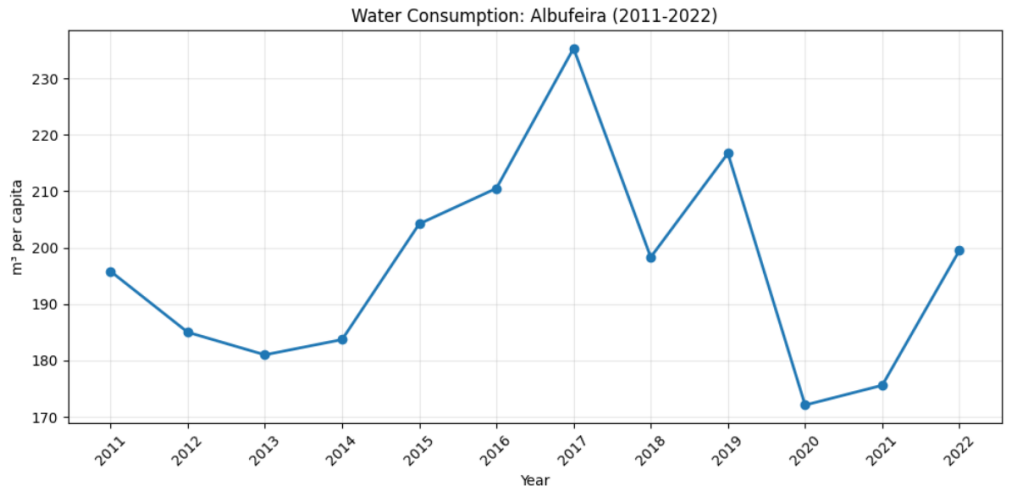
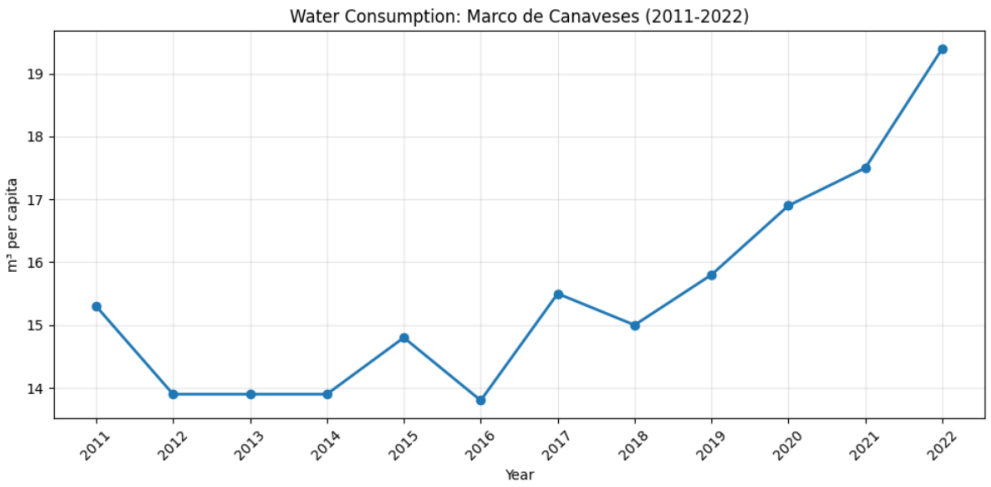
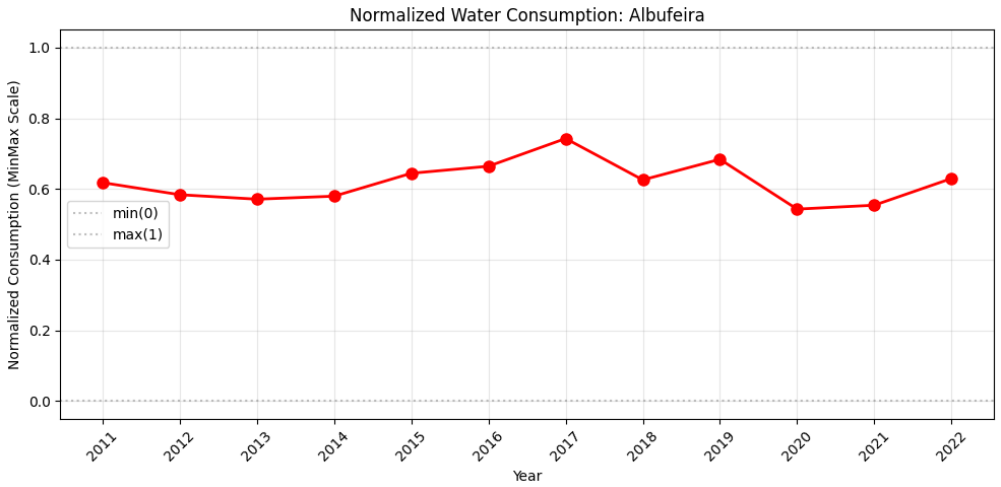
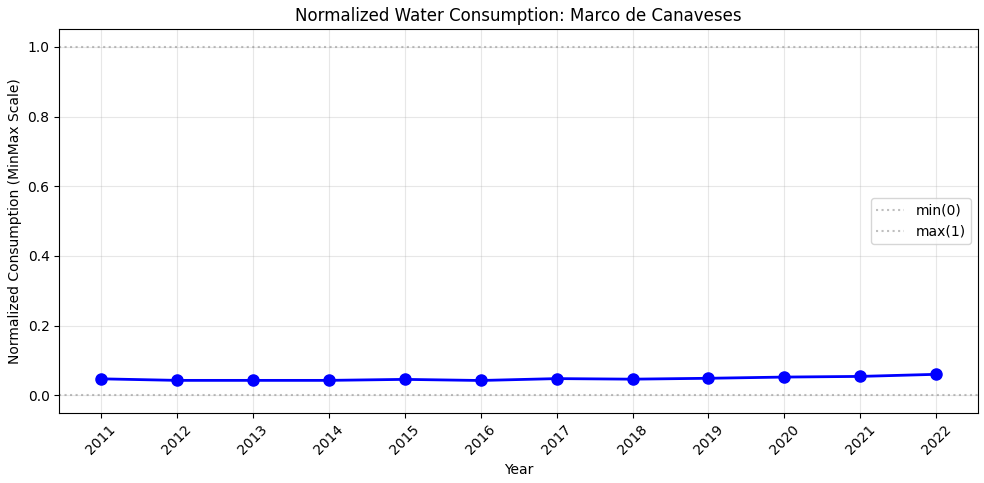








SPATIAL PATTERNS

- Random seed set
- Lattice size: 7x7 grid
  - 49 neurons
- U-Matrix: labels and dots
- Visual clusters

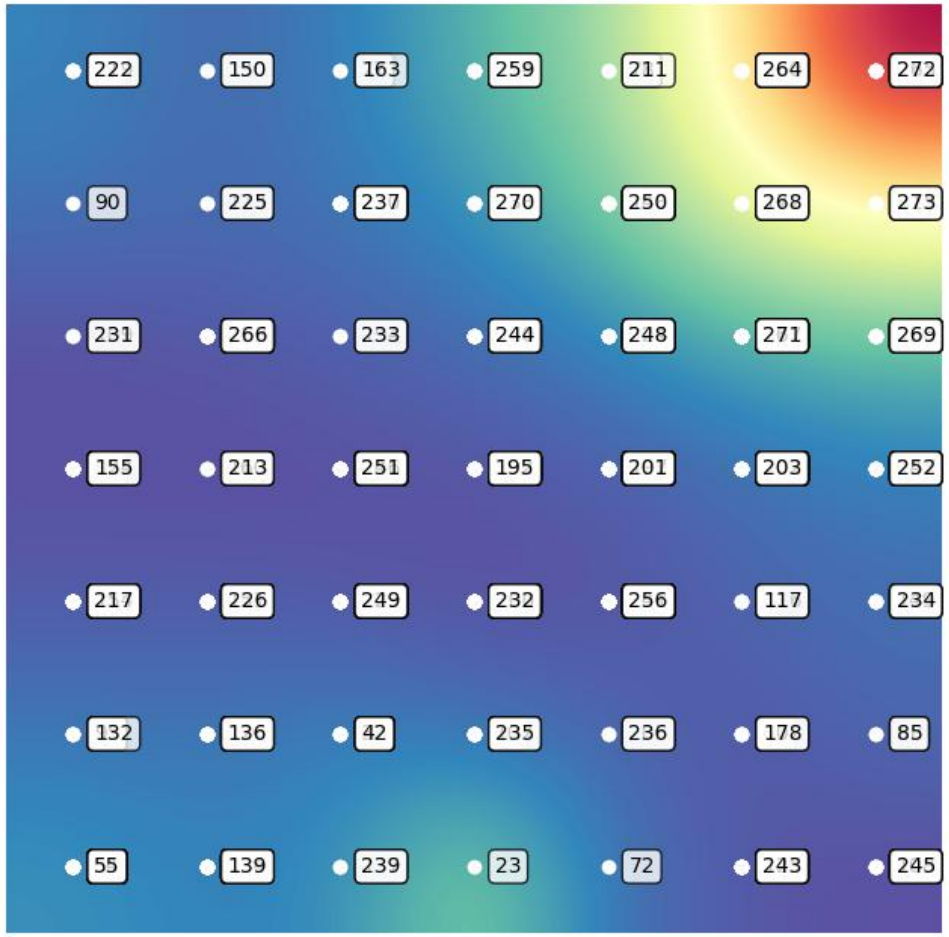
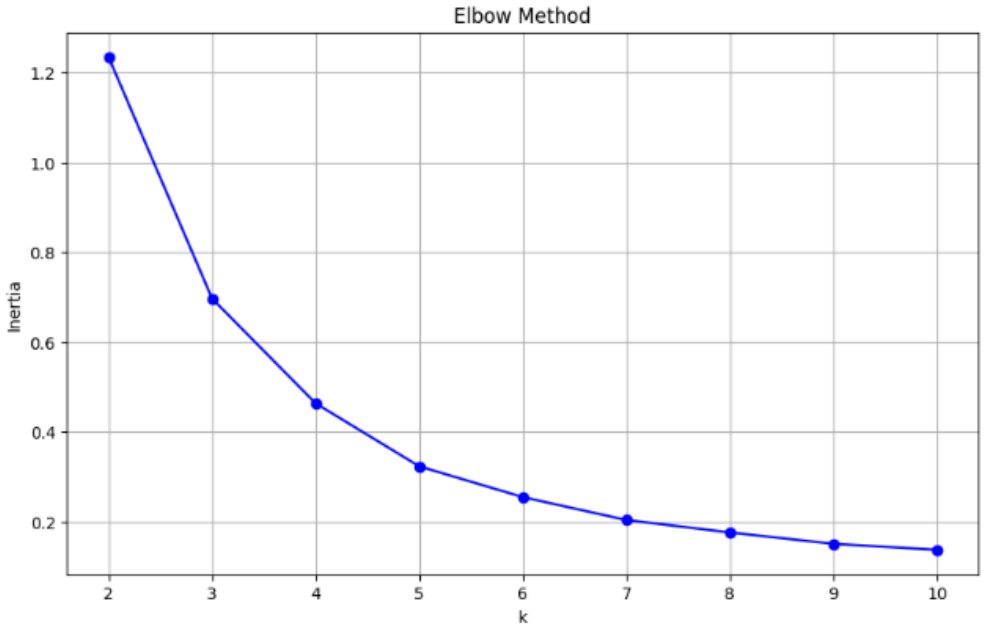


Figure 9 (right): U-Matrix

SECONDARY CLUSTERING



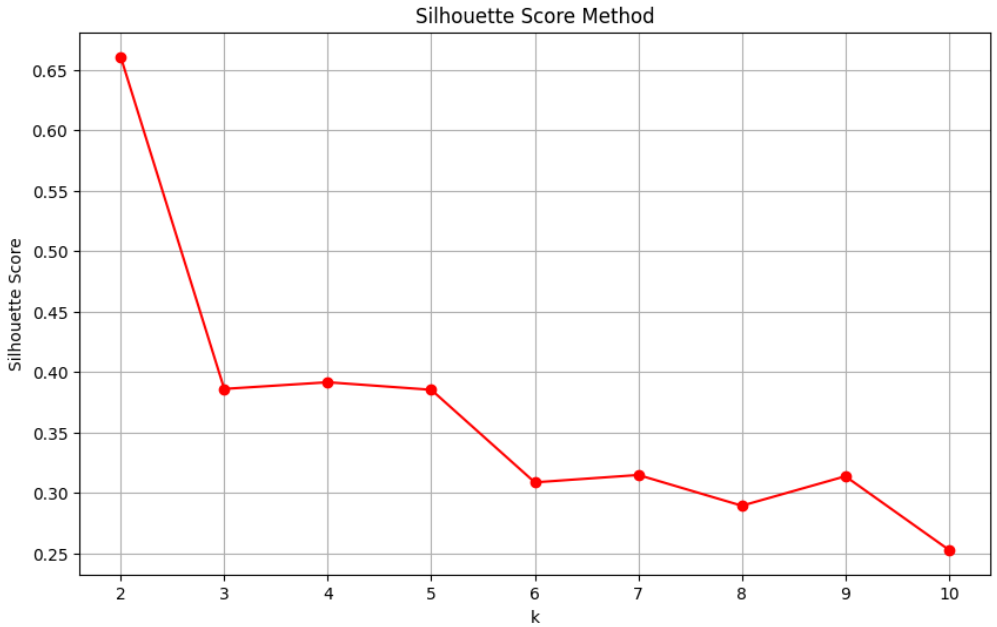


Figure 10 (left): Elbow Method; Figure 11 (right): Silhouette Score Method

SPATIAL CLUSTERING RESULTS

Cluster	Overall Average (m^3/hab)
0	36.6
4	55.8
1	73.1
3	115
2	169.5

Table 1 (above): Average raw water consumption per cluster (2011-2022)

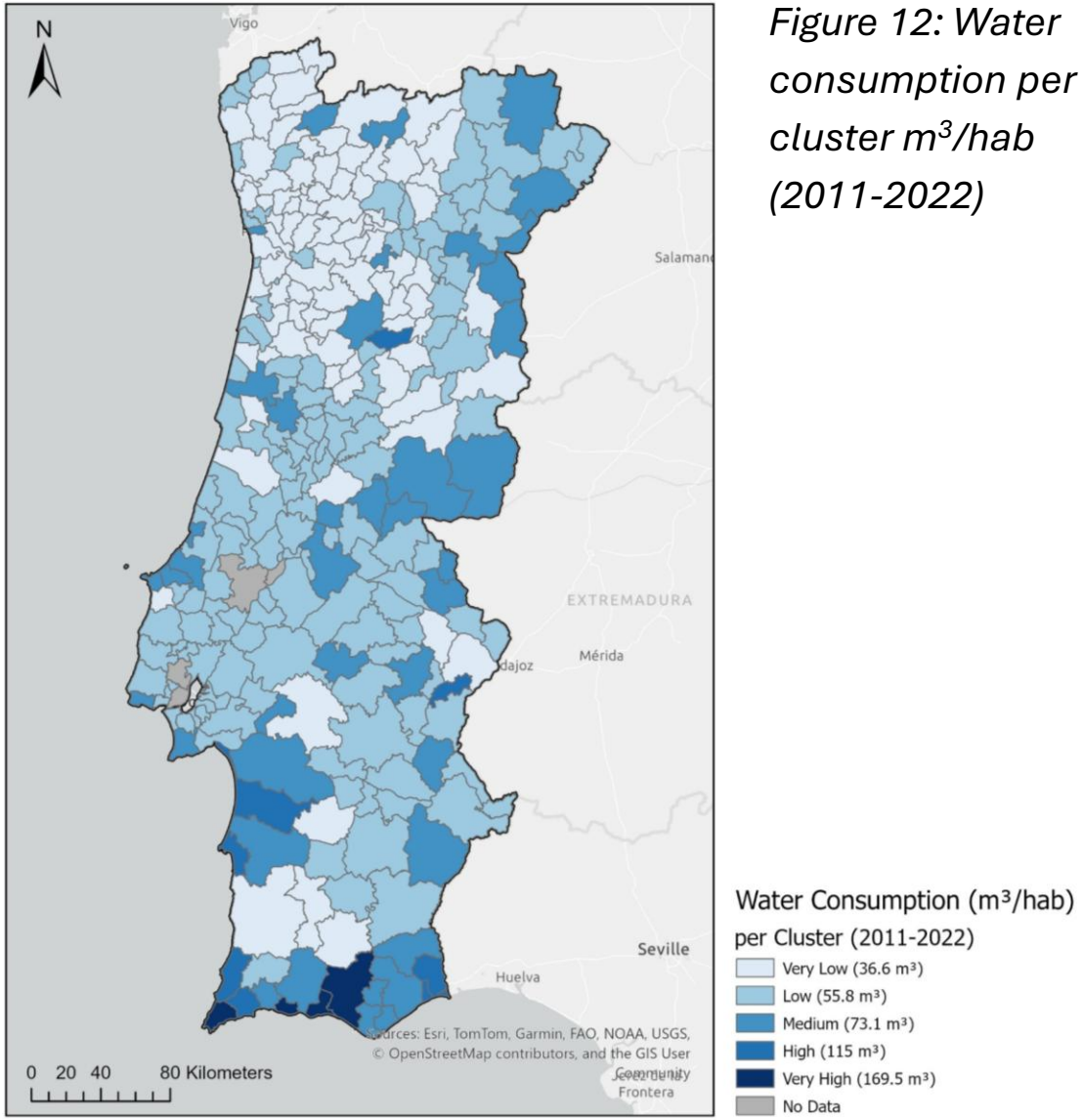
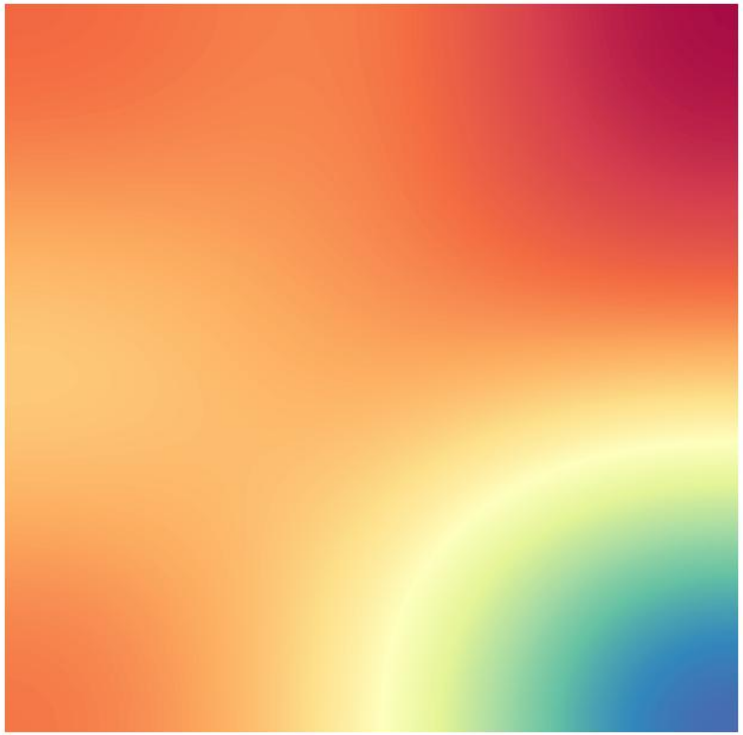


Figure 12: Water consumption per cluster m^3/hab (2011-2022)

TEMPORAL PATTERNS



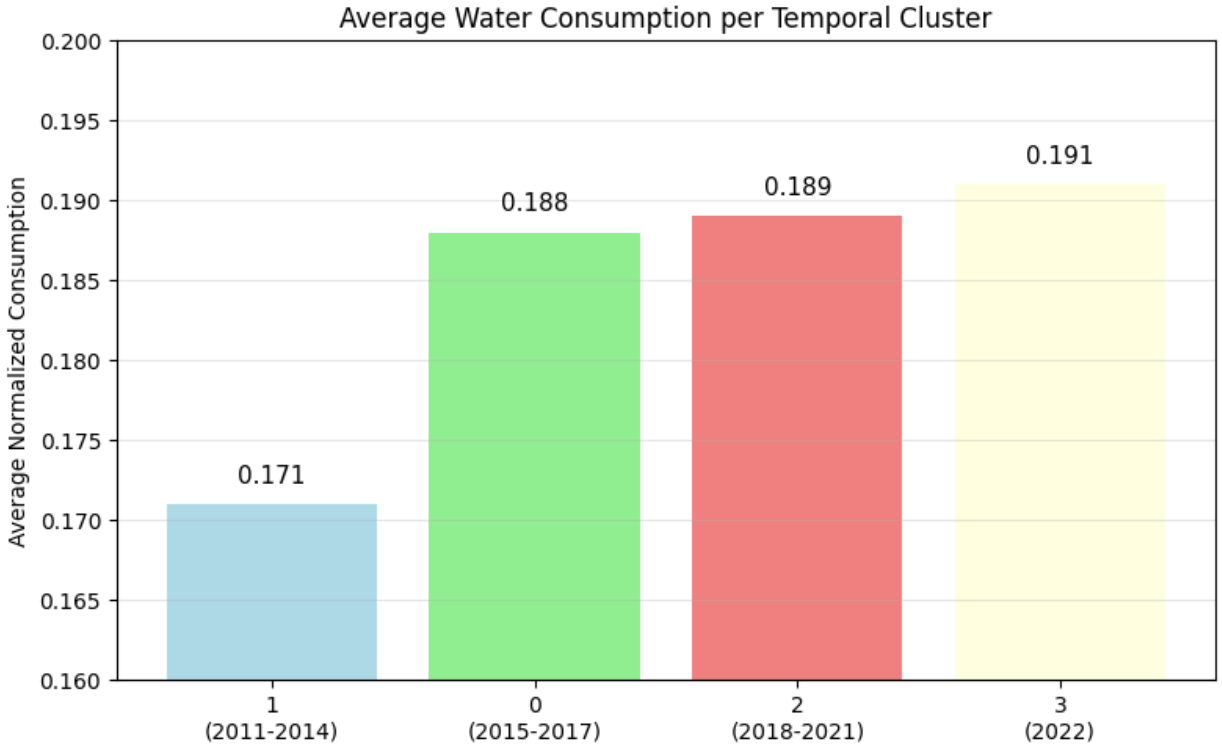


Figure 13 (left): U-Matrix; Figure 14 (right): Average Water Consumption per Temporal Cluster

TRAJECTORY ANALYSIS

- **Albufeira** (South, High consumption):
  - Volatile, responsive to each temporal regime
- **Cuba** (Alentejo, Medium consumption):
  - Stable, gradual increase
- **Marco de Canaveses** (North, Low consumption):
  - Flat trajectory, minimal response to national shocks

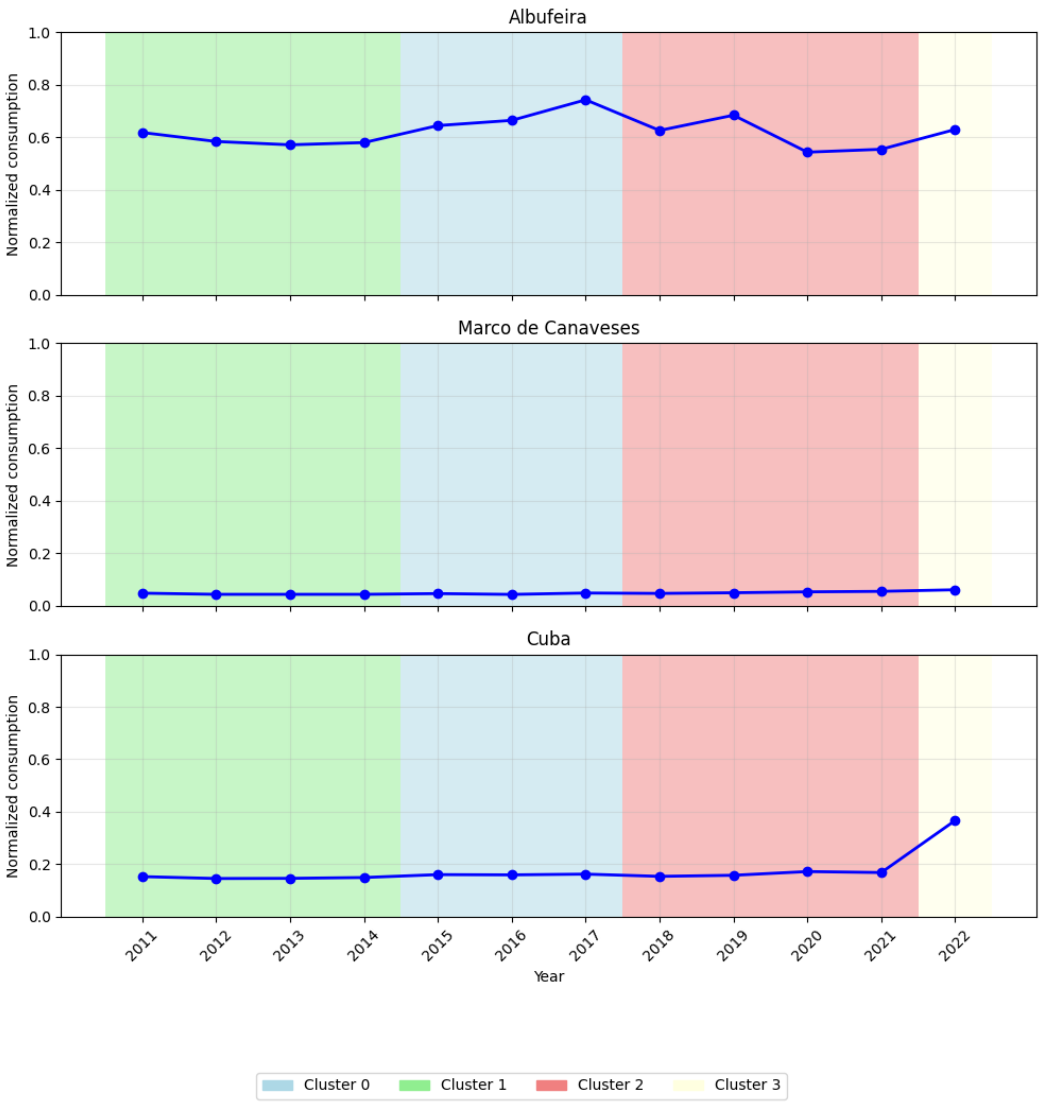
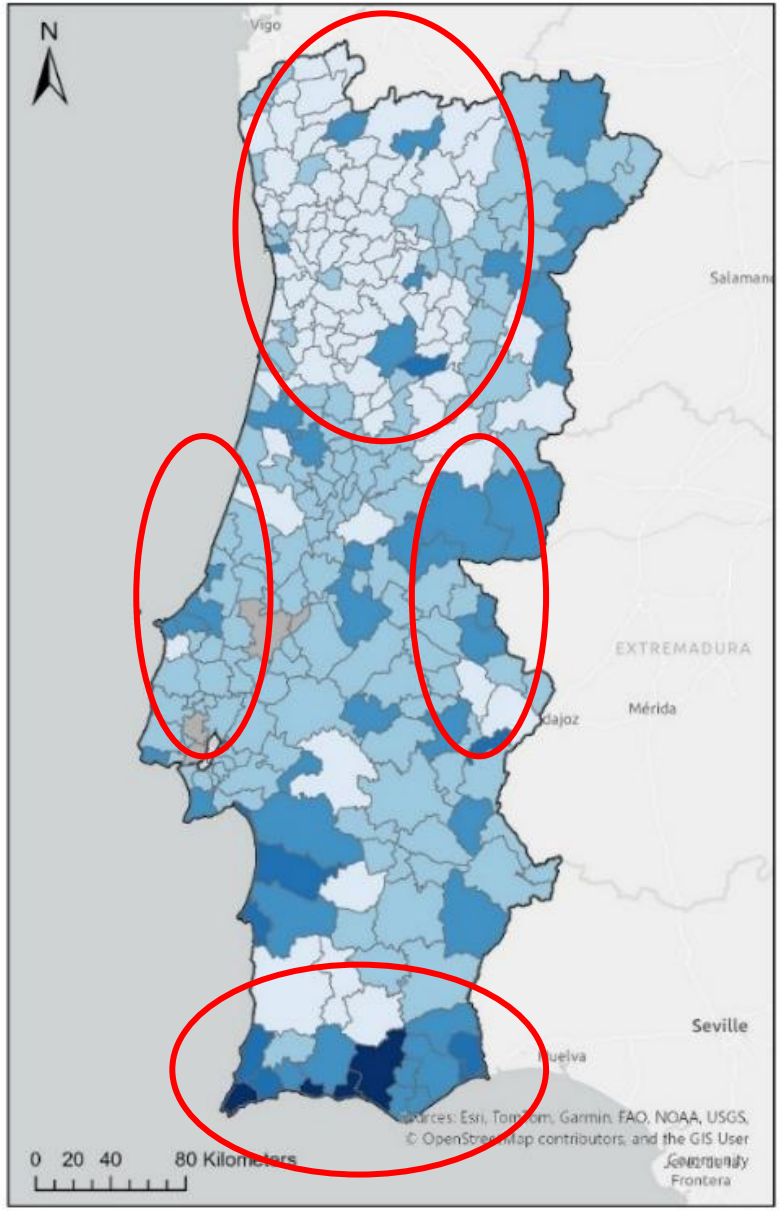


Figure 15: Normalized consumption rates for each municipality (Albufeira, Marco, and Cuba) overlayed onto temporal clusters

INTERPRETATIONS AND TAKEAWAYS

- Tourism-driven **south** (Cluster 2)
- Aging interior **north** (Cluster 0)
- Spanish **border** (Cluster 1 & 3) vs.

  Coast (Cluster 4)
- Overall **upward** trend



LIMITATIONS

- Excluded major municipalities
- "Per capita" excludes tourists, second-home owners
- Linear interpolation assumes linear trends
- Need for more time series across municipalities

FUTURE RESEARCH

Integrate tourism flow data

Add climatic variables (temperature, precipitation)

Include socioeconomic factors (income, age, housing type)

Seasonal (monthly) data would reveal intra-annual patterns

CONCLUSION

- SOMs: distinct regional water consumption patterns revealed
- Temporal analysis: four distinct periods
- Trajectory analysis: local context mediates national trends
- Need for regional water policies as well